



Original Article

Shorebird feeding specialists differ in how environmental conditions alter their foraging time

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Feeding specialization is a common cause of individual variation. Fitness payoffs of specialization vary with environmental conditions, but the underlying behavioral mechanisms are poorly understood. Such mechanistic knowledge, however, is crucial to reliably predict responses of heterogeneous populations to environmental change. We quantified spatiotemporal allocation of foraging behavior in wintering Eurasian oystercatchers (*Haematopus ostralegus*), a species in which feeding specialization can be inferred from bill shape. We combined global positioning system (GPS) and accelerometer data to quantify foraging time of 64 individuals for every tidal period in one or two winter seasons. Individuals varied widely in foraging time (3.7–6.5 h per tidal period) and individuals that spend more time foraging had lower inferred survival. Feeding specialization appeared a major determinant of individual variation in foraging time and its spatiotemporal allocation. Visually hunting worm specialists foraged more during day time and complemented intertidal foraging with grassland foraging when the exposure of intertidal flats was limited and nights were well illuminated. Shellfish specialists increased total foraging time in cold weather, whereas foraging time of worm specialists decreased as frosty grasslands became inaccessible. Our results imply that worm specialists may be most sensitive to cold snaps and daytime disturbance, whereas shellfish specialists are most sensitive to high water levels. These behavioral responses can be implemented in population models to predict the vulnerability of heterogeneous populations to environmental change and, thereby, provide a shortcut to long-term population studies that require fitness data across many years and conditions to make similar projections.

Key words: biologging, feeding specialization, foraging, *Haematopus ostralegus*, individual variation, movement ecology.

INTRODUCTION

Individuals vary in the expression of their mean phenotype (physiology, behavior, or demographic rate), as well as in their phenotypically plastic responses to environmental conditions. Understanding such variation is important for evolutionary and ecological dynamics. If individuals vary, and this phenotypic variation has a heritable basis and results in differential fitness, this may lead to an

evolutionary response. Furthermore, individual variation can affect population dynamics as it may cause that the mean performance of all individuals is different from the performance of an individual with the average phenotype (Bolnick et al. 2011).

One well-studied aspect in which individuals vary is feeding specialization (Durell 2000; Bolnick et al. 2003; Araújo et al. 2011; Ceia and Ramos 2015). Commonly, individuals specialize in different foraging techniques, for example, using different feeding methods to obtain the same food source, or in different diets, for example, by foraging on a limited selection of food sources (Annett and Pierotti 1999; Durell 2000; Bolnick et al. 2003). Feeding

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specialization may affect fitness through its effects on foraging efficiency (Terraube et al. 2014), foraging time (Cook et al. 2006), and mortality risk (Coleman and Wilson 1998; Darimont et al. 2007). Fitness consequences of specific feeding specializations, however, may vary depending on environmental conditions and available food sources as well as competition (Goss-Custard and Durell 1988; Beukema et al. 1993; Durell 2000; Darimont et al. 2007). Such fluctuating selection pressures may promote the maintenance of feeding specialization diversity (van de Pol et al. 2010).

Foraging time may be an important determinant of fitness associated with feeding specialization but how may depend on the situation. Feeding specializations that target profitable food sources, or use an efficient feeding technique, may need little time to meet their energy requirement and can, thus, shorten their foraging time (Araújo and Gonzaga 2007). However, restricted accessibility of a food source (e.g., in intertidal areas) might also limit foraging time (Durell 2000), the extent of which can vary with environmental conditions (e.g., fluctuating water levels). In such cases, foraging time is restricted to the period in which the food source is available. Consequently, individuals with the shortest foraging time are not necessarily the most efficient individuals.

Not only total foraging time but also the foraging time allocation may differ spatiotemporally among feeding specializations. For example, individuals with different feeding techniques might allocate their foraging time differently over the day. Individuals that exclusively search for prey by sight are less efficient in dark nights (Sitters 2000) and, consequently, it would be expected that those individuals allocate more foraging time to the day. An individual's feeding specialization may also influence where they forage as various food sources occur at different places in the landscape.

Comparisons of foraging time allocation of different feeding specializations under varying environmental conditions may reveal under which conditions individuals are most vulnerable. There are, however, very few studies that quantify foraging time differences among feeding specializations (Grantham et al. 1995) or even among-individual variation in foraging time (Kato et al. 2000; Daunt et al. 2014). This is likely due to the need to measure a substantial number of individuals with different feeding specializations over a prolonged period of time, representative of a wide range of environmental conditions, in all available feeding areas and during both day and night. This only recently became possible with advances in biologging and, specifically, the combined use of accelerometers (Yoda et al. 1999) and global positioning system (GPS) locations, enabling one to study animal behavior continuously by simultaneously quantifying time budgets and habitat usage (e.g., Ropert-Coudert et al. 2004; Watanabe et al. 2012; Brown et al. 2013; Daunt et al. 2014).

Here, we studied the consequences of feeding specialization for foraging time of 64 Eurasian oystercatchers (*Haematopus ostralegus*) using biologging for up to two nonbreeding seasons in the Dutch Wadden Sea estuary. The oystercatcher is a well-studied model system for the evolution and population level consequences of feeding specialization in the wild (Goss-Custard 1996). Oystercatchers show individual and sex-specific specialization in foraging behavior that results in a gradient of bill tips with different shapes (Norton-Griffiths 1967; Swennen et al. 1983; van de Pol et al. 2009). Feeding specialists range from worm and soft-shelled shellfish feeders (“worm specialists”), with a low bill tip height, to hard-shelled shellfish feeders (“shellfish specialists”), with a high bill tip height (Durell et al. 1993; van de Pol et al. 2009). Foraging habitat is known to differ among feeding specializations: although both

worm and shellfish specialists predominantly feed on intertidal flats, worm specialists also feed on inland grasslands (Goss-Custard and Durell 1983; Caldow et al. 1999; Sitters 2000; Durell et al. 2001).

Feeding specialization in oystercatchers may also have fluctuating fitness consequences because worm specialists had a lower survival during cold winters than shellfish specialists in the United Kingdom (Durell et al. 2001; Durell 2007), whereas a study from the Netherlands reported the opposite trend (van de Pol et al. 2010). A better understanding of the underlying mechanisms of how individuals with different feeding specializations adapt to different environmental conditions may improve our understanding of the variety of fitness patterns previously reported or may generate new testable hypotheses on how fitness payoff may vary spatiotemporally. However, these mechanisms are at the moment poorly understood. Therefore, we investigated the following four research questions: First, how much do individuals vary in their tidal foraging time in winter? Second, is foraging time associated with survival? Third, how much of this individual variation can be explained by feeding specialization? Finally, do birds with different feeding specializations allocate their foraging differently over time and space and does this make some foraging strategies more susceptible to environmental change? We will specifically explore how foraging time depends on 1) the exposure time of intertidal areas, 2) temperature, and 3) day–night patterns as this may inform us how individuals vary in their sensitivity to, respectively, global warming, sea-level rise, and disturbance (which mostly acts during daytime).

MATERIALS AND METHODS

Study area

The study was conducted in the western Dutch Wadden Sea estuary, comprising the islands Vlieland, Texel (northern half), and Griend and the surrounding intertidal flats (Figure 1). The majority of the study area consists of intertidal flats that are accessible to oystercatchers during low tide, whereas inland grasslands on Texel and Vlieland provide alternative foraging sites, for example, during high tide.

Data collection

We determined foraging time per tidal period (the period between two consecutive high-tide moments) by equipping birds with GPS trackers. Wintering oystercatchers were caught with mist nets on the western half of Vlieland in December 2016–January 2017 ($n = 20$) and December 2017 ($n = 42$), after which they were color-banded and equipped with GPS trackers. Birds in poor condition (<500 g) and birds with noticeable bill deformities (such as crossbill, gap in bill, and large asymmetry between upper and lower bills) were not equipped with a GPS tracker. None of the adult birds caught in winter were local breeding birds but bred in inland areas of the Netherlands and Germany, or in Scandinavia, which we confirmed by visually inspecting GPS tracks of the breeding season. Additionally, 20 local breeding oystercatchers were caught on the nest and equipped with trackers between May and July 2017. Eleven trackers did not generate any data (individual bird details in [Supplementary Table S1](#)).

In all analyses, we used bill tip height (measured 3 mm from the bill tip using a caliper) as a continuous proxy for the type of feeding specialization of an individual (van de Pol et al. 2009). For presenting our results in figures we used categories for visual reasons

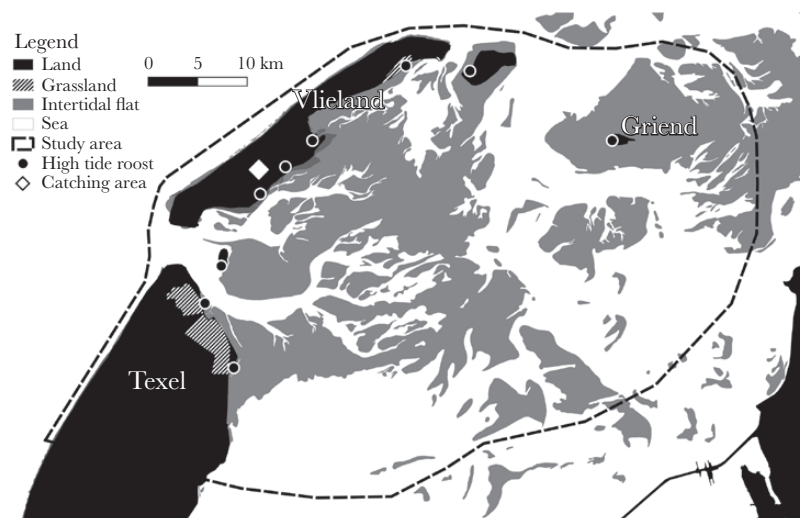


Figure 1

Study area including the Wadden islands Vlieland and Texel, intertidal flats, and grasslands (note the small grassland area on Vlieland). On Texel only grasslands are indicated which were used by oystercatchers.

only: worm specialists (pointed bill shape), Intermediate (pointed chisel bill shape), and shellfish specialists (chisel or blunt bill shape). The corresponding bill shapes were determined in the field based on bill tip height, width, and shape (van de Pol et al. 2009) and corresponded with bill tip height ranges of 2.3–3.3, 3.0–4.9, and 4.3–5.7 mm for pointed, pointed chisel, and chisel or blunt bill shapes, respectively. The sex of a bird was determined by DNA analysis of a small blood sample taken from the wing vein. Age class (juvenile, second winter or adult) was determined based on morphology (Cramp et al. 1983). Standard biometry measurements were taken, of which tarsus-toe length was used as a proxy for body size.

The UvA-BiTS GPS trackers we used collected data on both location and behavior of birds (Bouten et al. 2013). The tracker (13.5 g) included a rechargeable battery, solar panel, and accelerometer and was attached on the back using a Teflon harness (2 g), in total weighing 15.5 g (2.00–3.14% of the mass of the oystercatchers; mass range 494–774 g; Figure 2a). The trackers sampled GPS positions in 5–60-min intervals. A 0.35-s accelerometer sample (20 Hz; triaxial measuring surge X, sway Y, and heave Z) was measured at least every 10 min, independent of the GPS interval. GPS positions were used to determine the foraging habitat and accelerometer samples were used to classify behavior (see below). When the battery of the tracker was empty, which happened most often between November and January, no GPS or accelerometer measurements were taken until it was recharged. All data collected by the tracker were stored on internal memory and downloaded via a radio antennae network (Bouten et al. 2013) permanently present on Vlieland during the study period. Ring readings in the study area were done throughout the year on the islands Vlieland and Texel to detect individuals with a defective tracking device. We used ring readings (registered by us and volunteers on www.wadertrack.nl) in combination with detection by the antenna system to determine apparent survival.

We annotated accelerometer data using video recordings and used a machine learning approach to classify accelerometer samples into distinct behaviors (Shamoun-Baranes et al. 2012, 2016). GPS-equipped oystercatchers were filmed on Vlieland from March to July 2017, when the tracker batteries were fully charged. The GPS tracker settings were adjusted before the filming events to

record 10-s 20-Hz accelerometer samples (i.e., 200 accelerometer measurements) in 20–60-s intervals (Figure 2b). Accelerometer data were synchronized with the recordings and then annotated into five behaviors (Supplementary Table S2): flying, walking, foraging, preening, and inactive. Extra flight behavior samples were selected based on GPS speed (speed >10m/s) and visual inspection of the accelerometer data (see Supplementary Table S2; see how flight behavior can be easily recognized) to increase the number of flight behaviors in the annotated accelerometer data set. Washing and aggressive behavior were not included in the classification under the assumption that these behaviors occurred very rarely during winter (even during the postwinter observations and filming, washing and aggressive behavior were rarely observed).

The annotated accelerometer data were subsequently used to train a random forest algorithm (Breiman 2001) using summary features of the accelerometer samples (following Shamoun-Baranes et al. 2016). The annotated accelerometer data were split into 0.35-s samples (seven accelerometer measurements), yielding in total 33 107 annotated samples. The samples were divided into training (24 929 samples) and testing (8178 samples, individuals other than in training data set) data sets to construct the classification models (Supplementary Table S3). We calculated 33 statistical summary features from the accelerometer samples, which are described in Shamoun-Baranes et al. (2016; see also Supplementary Table S4). The most relevant features were selected by constructing 10 decision trees using C4.5 machine learning algorithm (Quinlan 1993) using 300 random samples of each behavior from the training data set. The features that occurred in the top four nodes in any of the trees were retained, whereas the other features were excluded. This process was done three times (Shamoun-Baranes et al. 2016). The 15 retained features (Supplementary Table S4) were used to train a random forest with 50 trees using 1716 samples (the minimum available for all behaviors) for all five behaviors.

The testing dataset was used to calculate the prediction accuracy of the random forest model. Prediction accuracy was 94.6% ($n = 8.178$; $\kappa = 0.927$; Supplementary Table S5) and especially high for foraging, flying, inactive, and preening behavior. Walking was sometimes confused with foraging as prey searching by sight (foraging) is identical to a slow walking behavior and even in the

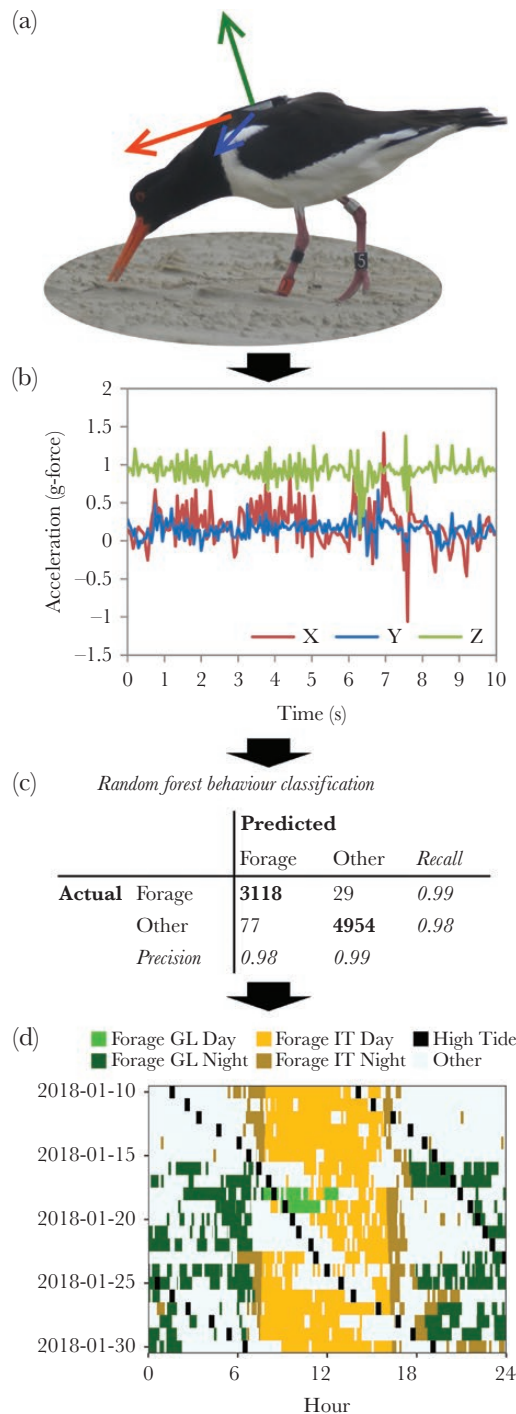


Figure 2
(a) Oystercatcher with UvA-BiTS GPS tracker showing triaxial acceleration (X: red, Y: blue, Z: green). (b) Example of a 10-s acceleration sample of a foraging bird obtained for calibration of the accelerometer. (c) Random forest behavior classification accuracy, separating foraging (including walking) behavior from other behaviors. (d) Example of a time budget of one individual (intermediate specialization) showing intertidal (IT) and grassland (GL) foraging during day and night. Tidal foraging time was calculated between two consecutive high tides (highlighted in black).

field not always distinguishable. True walking behavior, however, occurs often during the breeding season when oystercatchers tend to walk a lot inside their territory. Because our study focused on the

nonbreeding season, we assumed that predicted walking behavior in winter was actually most often foraging behavior. Indeed, the pattern of foraging and walking behaviors over the tidal period was very similar (Supplementary Figure S1), which would not be the case if walking often occurred during high tide (e.g., at high-tide roost sites). We, therefore, combined foraging and walking behavior into one foraging class. In this study, we focused on foraging time (compared with other behaviors); our classification model distinguished foraging (including walking) from other behaviors correctly in 98.7% of all cases ($n = 8178$; $\kappa = 0.973$; Figure 2c).

Based on GPS locations and behavior classifications, intertidal and grassland foraging time (hours) was calculated for every bird for every tidal period (Figure 2d). A tidal period was defined as the period between two consecutive high-tide moments at Vlieland Haven (Rijkswaterstaat 2018); thus, each tidal period (~ 12.4 h) covered a single low tide. All behavior measurements within a tidal period were weighed based on the interval between consecutive measurements, and the intervals of foraging behaviors were summed to acquire one total foraging time estimate per tidal period.

In winter, GPS fixes were taken with hourly intervals to save battery and extra behavior measurements were taken in between with 10-min intervals. Consequently, locations for behavior measurements were then derived from the closest GPS fix. This did, probably, not strongly affect estimates of intertidal and grassland foraging time as oystercatchers did not frequently switch between the foraging habitats. Using only the GPS fixes collected with 5-min intervals, we estimated that switching oystercatchers stayed on average for 21.8 consecutive h in intertidal areas and 11.3 consecutive h (approximately one tidal period) in grasslands.

To correct for environmental variables, daily weather data was obtained from the Royal Netherlands Meteorological Institute weather station at Vlieland (KNMI 2018). Daily moon illumination values were obtained via the *lunar.illumination* function in the *lunar* package in R.

Statistical analysis

We focused our analyses on the nonbreeding season (1 August to 31 March). We omitted data if 1) it concerned the first five tidal periods after the oystercatcher was equipped with a GPS tracker ($n = 279$ tidal foraging time estimates), 2) there were less than 70 behavior measurements in the tidal period ($n = 459$), 3) the maximum interval between two behavior measurements was more than 20 min ($n = 537$), or 4) one or more GPS positions were outside the study area (Figure 1; $n = 769$). We subsequently removed seven birds from the analysis that had fewer than 30 tidal foraging time estimates ($n = 95$). In total, we retained 8704 tidal foraging time estimates from 661 tidal periods and 64 birds (15, 29, and 20 worm, intermediate, and shellfish specialists, respectively; Supplementary Table S1).

We analyzed both intertidal foraging time and total (intertidal + grassland) foraging time as a response variable. First, we examined whether foraging time was related to survival. Then, we examined which environmental factors influenced foraging time and quantified among-individual variation in foraging time. Then, we examined how much of the among-individual variation could be explained by individual traits, especially feeding specialization, and by different foraging strategies of individuals with different feeding specializations. All analyses were performed in R 3.3.2 (R Core Team 2017).

For birds caught in winter 2016–2017 and summer 2017, we investigated whether foraging time was associated with apparent

survival using binary logistic regression based on whether oyster-catchers were recorded (either by GPS tracker or ring reading in the field) 1 year after release (Supplementary Table S1).

We used linear mixed-effects modeling to correct for environmental factors that affect intertidal and total foraging time per tidal period. Intertidal and total tidal foraging times (in hours) were separately used as response variables. Bird ID was included as random intercept. The following variables were included as explanatory variables: *Winter* (2016–2017 or 2017–2018), *Proportion night* (0–1, where 1 indicates that the whole tidal period was between sunset and sunrise), *Moon illumination* (0–1, where 1 is full moon) and the linear and squared terms of *Seasonday* (1 August = day 1), *Temperature* (mean daily temperature in °C), *Windspeed* (mean daily windspeed in kilometers per hour), *Proportion intertidal flat exposure* (proportion of time when water level < -10 cm Amsterdam Ordnance Datum [NAP]), *Proportion intertidal flat exposure previous 10 tides* (Supplementary Table S6). We included the intertidal flat exposure over the previous 10 tides to determine whether birds compensated in foraging time after prolonged periods of high water levels. The variables *Seasonday*, *Temperature*, and *Windspeed* were z-transformed prior to analysis. We used an information theoretic model selection approach using Akaike's Information Criterion (AIC; Burnham and Anderson 2004). Models with all possible combinations of predictor variables were run and the model with the lowest AIC was selected for both the intertidal and total foraging time and used in further analyses. This approach yielded the same environmental base models as when we used backward selection based on *P* values, indicating that the environmental base models were not depending on the selection procedure. Full model details and *P* values of all variables are provided in Supplementary Table S7. The residual distributions of the environmental base models (one for each of the two response variables) were visually checked for normality (Supplementary Figure S2). The variance estimate of Bird ID as random intercept provided an estimate of the among-individual variation in foraging time.

Subsequently, we quantified how much the following five individual traits explained of the among-individual variation in intertidal or total foraging time by adding them to the environmental base models: 1) *Bill tip height* (millimeter), 2) *Sex* (Male or Female), (3) *Age class* (Juvenile (first winter), Subadult (second winter) or Adult (third winter and older; birds can age over seasons), (4) *Tarsus-toe length* (millimeter; measure for bird size), and (5) *Breeding site* (local breeder on Vlieland or nonlocal breeder; Supplementary Table S6). Z-transformation of *Tarsus-toe length* and *Bill tip height* was applied prior to analysis. Models were constructed in which each of the traits was added singly and where all traits were added simultaneously to the environmental base models. We tested for significance of fixed terms using Wald chi-square tests by comparing models with one dropped term against the full model for the model including all individual traits. We compared the marginal R^2 (R^2_m , explained variation of fixed effects) and the conditional R^2 (R^2_c , explained variation of fixed and random effects; Nakagawa and Schielzeth 2013) of all constructed models with the environmental base models to quantify the explained among-individual variation in foraging time by individual traits. We calculated 1) additional variation explained by fixed effects (R^2_m model – R^2_m base model), 2) individual variation not explained by individual traits (R^2_c model – R^2_m model), and 3) individual variation explained by adding individual traits as explanatory variables ($[R^2_c \text{ base model} - R^2_m \text{ base model}] - [R^2_c \text{ model} - R^2_m \text{ model}]$).

To determine whether individuals with different feeding specializations differed in foraging time allocation, we constructed six

models for intertidal and total foraging time, including interactions between bill tip height and environmental factors: *Bill tip height* in interaction with 1) *Proportion intertidal flat exposure*, 2) *Proportion intertidal flat exposure previous 10 tides*, 3) *Temperature*, 4) *Proportion night*, 5) *Proportion night and Moon illumination* (three-way), and 6) all interactions mentioned before. We tested for significance of fixed terms using Wald chi-square tests by comparing models with one dropped term against the full model for the model including all interactions. We calculated additional variation explained by the fixed effects and individual variation not explained by individual traits based on R^2_m and R^2_c values (see above).

RESULTS

Individual variation in foraging time

Mean individual foraging time ranged between 3.66 and 6.51 h per tidal period (lower quantile = 4.18 h, median = 4.62 h, upper quantile = 5.14 h; Figure 3a). A total of 16.5% and 16.9% of variation

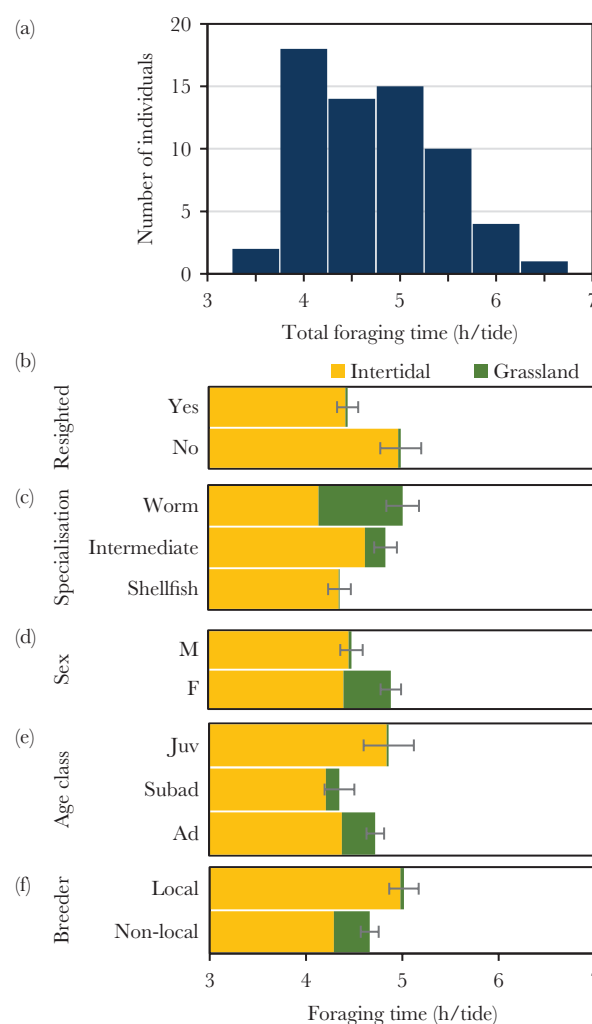


Figure 3

(a) Histogram of mean total (intertidal + grassland) foraging time per individual. Mean intertidal and grassland foraging time $\pm$ standard error (SE) for total foraging time for (b) birds that were resighted or not resighted 1 year after the catch ($n = 29$), (c) different specializations ($n = 64$), (d) sex ($n = 64$), (e) age class ($n = 64$), and (f) local or nonlocal breeding sites ($n = 64$).

in intertidal (standard deviation [SD] among individuals = 0.64 h) and total foraging time (SD = 0.62 h) could be attributed to among-individual variation, respectively (Table 1). There was no indication that the amount of individual variation was inflated by sampling variance (i.e., individual variability was independent of sample size; Supplementary Figure S3). All environmental variables except *Winter* were retained in the environmental base models for intertidal and total foraging time when estimating the amount of individual variation (Table 2).

Foraging time and apparent survival

Birds that were not recorded 1 year after they were released ($n = 9$) had a longer intertidal foraging time ($\chi^2 = 6.2$, degrees of freedom [df] = 1, $P = 0.013$) and total foraging time ($\chi^2 = 5.8$, df = 1, $P = 0.016$) than those that were observed ($n = 20$; Figure 3b).

Individual traits determining foraging time

Bill tip height explained 21.9% of the individual variation in total foraging time (3.7% out of 16.9% variation that could be attributed to among-individual differences; Table 1). Bill tip height significantly explained variation in total foraging time but not intertidal foraging time (Table 2; Figure 3c). Birds with a lower bill tip height foraged longer, which could be attributed to grassland foraging (Figure 3c). Intertidal foraging time was longer in young birds and local breeders but there was no variation between sexes (Table 2; Figure 3d–f). Breeding site explained 17.0% of variation of the individual variation in intertidal foraging time (2.8% out of 16.5%; Table 1). Although sex by itself explained 7.7% of individual variation in total foraging time (1.3% out of 16.9%; Table 1), it was not significant in the model with all individual traits (Table 2) as sex and bill tip height were highly correlated ($r = 0.65$). The difference between local and nonlocal breeders was less pronounced in total foraging time as nonlocal breeders foraged longer on grasslands (Table 2; Figure 3f). All five individual traits combined accounted

for 26.1% (4.3% out of 16.5%) and 30.7% (5.2% out of 16.9%) of the among-individual variation for intertidal and total foraging time, respectively.

Differential responses of feeding specializations to environmental conditions

Feeding specialists showed clear differences in foraging time responses to three environmental conditions (intertidal flat exposure, temperature, and day and night) (Figure 4). During relatively high low tides (e.g., during bad weather) intertidal feeding areas were exposed for a shorter period than normal. Shellfish specialists' (high bill tip height) total foraging time was lower during tides with low intertidal exposure, whereas worm specialists (low bill tip height) slightly increased their total foraging time by grassland foraging (Figure 4b; Table 3). Worm specialists foraged less on intertidal flats and more on grasslands when intertidal flat accessibility was limited over prolonged periods (Figure 4c; Table 3).

On cold days, shellfish feeders increased their total foraging time, whereas foraging time of worm feeders decreased (Figure 4d; Table 3). For all individuals, intertidal and total foraging time was longest during daytime tides (Table 2; Figure 4e). Worm specialists foraged longer than shellfish specialists during daytime tides (Figure 4e; Table 3). During the night, foraging time was similar among feeding specializations around new moon periods but, around full moon, worm feeders foraged longer, which could mainly be attributed to grassland foraging (Figure 4e; Table 3).

More (6.5%) variation in total foraging time was explained by fixed variables in the model that included bill tip height in interaction with night, intertidal flat exposure, and temperature than in the base model with only environmental variables (Table 1). Feeding specialization in interaction with environmental variables explained 26.6% of among-individual variation in total foraging time (variation explained by random effect was reduced from 16.9% to

Table 1
Variation in foraging time explained by environmental base model, individual traits, and by interactions of bill tip height with environmental variables. The table provides estimates of additional variation explained by fixed factors in comparison to the environmental base models (Δ variation explained), explained individual variation by individual traits (Explained ind. var.), individual variation that remains unexplained (Unexplained ind. var.; see methods for calculations) and the marginal R^2 (R^2_m) and conditional R^2 (R^2_c). Explained individual variation was not calculated for models with interactions as R^2 was affected by both environmental variables and bill tip height. bth = bill tip height

Model	Intertidal foraging time				Total foraging time				
	Δ variation explained	Explained ind. var.	Unexplained ind. var.	R^2_m	R^2_c	Δ variation explained	Explained ind. var.	Unexplained ind. var.	R^2_m R^2_c
Individual traits	Environment (base model)		16.5%	0.268	0.432			16.9%	0.218 0.387
	Environment + bth	−0.2%	−0.5%	16.9%	0.266	0.435	3.8%	3.7%	0.257 0.389
	Environment + sex	−0.1%	−0.4%	16.9%	0.267	0.435	1.6%	1.3%	0.234 0.387
	Environment + age class	0.5%	0.7%	15.8%	0.273	0.430	0.1%	0.2%	0.219 0.387
	Environment + tarsus-toe	0.3%	0.0%	16.5%	0.271	0.436	0.1%	−0.1%	0.219 0.389
	Environment + breeding site	3.2%	2.8%	13.7%	0.299	0.436	1.0%	0.9%	0.229 0.389
	Environment + all individual traits	4.3%	4.3%	12.2%	0.311	0.432	5.4%	5.2%	0.273 0.390
Environment responses	Environment + bth × exposure	0.2%		17.1%	0.270	0.441	4.8%	13.1%	0.267 0.397
	Environment + bth × 10-tide exposure	0.5%		16.9%	0.273	0.442	4.8%	12.9%	0.266 0.395
	Environment + bth × temp	−0.1%		16.8%	0.266	0.435	5.6%	12.3%	0.274 0.397
	Environment + bth × night	1.8%		15.9%	0.286	0.445	4.1%	13.1%	0.259 0.390
	Environment + bth × night × moon	1.9%		15.9%	0.287	0.446	4.4%	13.1%	0.263 0.394
	Environment + all bth interactions	3.0%		15.0%	0.298	0.448	6.5%	12.4%	0.283 0.407

Table 2

Linear mixed model for intertidal foraging time ($R^2 = 0.432$) and total foraging time ($R^2 = 0.390$) per tidal period in hours using environmental and individual traits as explanatory variables. Environmental variables were selected based on AIC comparison of models with all possible combinations of environmental variables and using bird individual as a random intercept. Reference categories for sex, age class, and breeding sites were female, adult, and nonlocal breeder, respectively. Seasonday, temperature, windspeed, tarsus-toe, and bill tip height were z-transformed. Variance (σ^2) of the random effect, bird individual ($n = 64$), was 0.307 and 0.272 for the intertidal and total foraging time models, respectively

		Intertidal foraging time					Total foraging time				
		Estimate	SE	df	χ^2	sig.	Estimate	SE	df	χ^2	Sig.
Environment	Intercept	3.18	0.23				4.76	0.22			
	Winter	—					—				
	Seasonday	−0.17	0.03	1	34.7	***	−0.15	0.03	1	26.9	***
	Seasonday ²	−0.18	0.02	1	57.3	***	−0.22	0.02	1	79.6	***
	Temperature	−0.23	0.04	1	32.4	***	−0.18	0.04	1	21.4	***
	Temperature ²	—					—				
	Windspeed	0.13	0.02	1	54.1	***	0.19	0.02	1	120.1	***
	Windspeed ²	—					0.05	0.01	1	17.9	***
	Exposure	7.07	0.33	1	400.9	***	6.72	0.35	1	357.7	***
	Exposure ²	−4.87	0.38	1	146.7	***	−5.66	0.41	1	191.0	***
	10-tide exposure	6.20	0.79	1	60.0	***	2.84	0.78	1	13.1	***
	10-tide exposure ²	−7.90	0.84	1	84.4	***	−5.44	0.84	1	41.7	***
	Night	−3.32	0.08				−3.42	0.08			
	Moon	−0.73	0.08				−0.96	0.08			
Individual traits	Night × moon	1.13	0.12	1	88.5	***	2.07	0.12	1	295.7	***
	Bill tip height	−0.02	0.09	1	0.1	ns	−0.27	0.08	1	10.5	**
	Sex (male)	−0.03	0.19	1	0.0	ns	−0.08	0.18	1	0.24	ns
	Age class			2	8.6	*			2	5.0	ns
	Juvenile	0.38	0.13				0.26	0.13			
	Subadult	0.14	0.09				0.06	0.09			
	Tarsus-toe	0.12	0.07	1	1.6	ns	0.08	0.07	1	1.6	ns
	Breeding site (local)	0.79	0.19	1	4.2	***	0.42	0.17	1	6.0	*

Sig. = *0.05 > P > 0.01, **0.01 > P > 0.001, *** P < 0.001.

12.4%; Table 1). Environmental variables and feeding specialization together explained 44.8% and 40.7% of variation in intertidal and total foraging time, respectively (Table 1).

DISCUSSION

Using combined GPS and accelerometer measurements, we quantified foraging time and its spatiotemporal allocation over a wide range of environmental conditions for a substantial number (64) of individuals. We found considerable among-individual variation in foraging time. Individuals that foraged longer were less likely to be recorded again 1 year after they were equipped with a GPS tracker, implying a lower survival. Feeding specialization explained a considerable part (21.9%) of the among-individual variation in total foraging time and could account for sex differences in foraging time. Most interestingly, our results show that birds with different feeding specializations alter their foraging time differently under varying environmental conditions, suggesting they differ in their susceptibility to environmental change. Specifically, worm specialists complemented intertidal with grassland foraging when the intertidal flats were less exposed, shortened instead of lengthened their foraging time during cold snaps, and allocated more foraging time to the day.

Fitness consequences of foraging time

Individuals who forage longer under normal conditions (because of lower foraging efficiency or higher energy expenditure) are expected to be more prone to deteriorating body condition and

mortality when available foraging time becomes limiting. Although foraging time of an individual can decrease through exhaustion in the period directly prior to death (Daunt et al. 2007), weak or inefficient individuals are known to forage longer to maintain or regain their body condition (Powolny et al. 2015). This is indeed supported by our finding of lower apparent survival for individuals with long foraging times. In 1 or 2 years, we should have sufficient data to more accurately estimate the effect of foraging time on true (rather than apparent) survival using capture-mark-resighting modeling on all 64 individuals.

Increased foraging time is not necessarily disadvantageous when individuals specialize in food sources that are available at different spatiotemporal scales. The intake payoffs of feeding specializations vary depending on the environmental conditions and population structure (Svanbäck and Persson 2009; van de Pol et al. 2010). For example, specializing in a less profitable food source that is accessible for a larger proportion of time may be the best strategy when the accessibility of a more profitable food source is limited. In oystercatchers, grasslands are traditionally regarded as less profitable and, consequently, a refuge feeding habitat for weaker individuals that cannot meet their energy requirements on intertidal flats (Goss-Custard and Durell 1983; Caldow et al. 1999; Durell et al. 2001). However, worm specialists sometimes choose to feed on inland grasslands during low tide when shellfish specialists forage on intertidal flats (Figure 4a; Heppleston 1971). This suggests that grasslands cannot exclusively be regarded as a supplementary foraging habitat as, under some circumstances, grassland foraging appears a preferred choice of worm specialists.

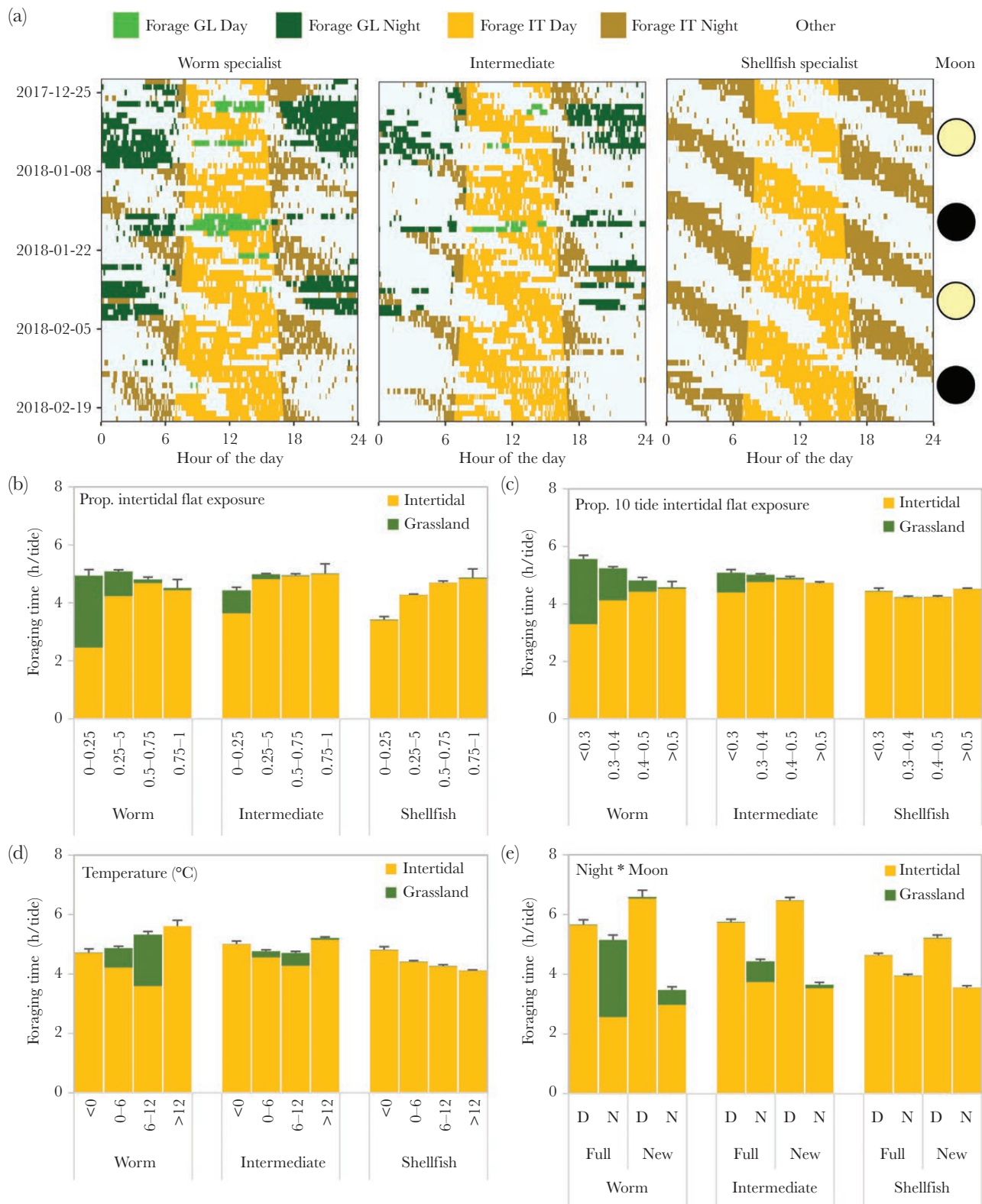


Figure 4

(a) Time budget examples of oystercatchers with different specializations showing intertidal (IT) and grassland (GL) foraging during day and night. Intertidal and grassland foraging time $\pm$ standard error (SE) for total foraging time for different feeding specializations for (b) different intertidal flat exposure proportions (proportion low water), (c) average intertidal flat exposure proportions over the previous 10 tidal periods, (d) different temperatures, (e) daytime (D, proportion night = 0–0.25) and nighttime (N, proportion night = 0.75–1) tidal periods in full (moon illumination = 0.75–1) and new moon (moon illumination = 0–0.25) periods. For significance of effects, see Table 3.

Table 3

Linear mixed model for intertidal foraging time ($R^2 = 0.448$) and total foraging time ($R^2 = 0.407$) per tidal period in hours using environmental factors and feeding specialization as explanatory variables, including their interaction. Environmental factors were selected based on AIC comparison of models with all possible combinations of environmental factors and using bird individual as random intercept. Seasonday, temperature, windspeed, and bill tip height (bth) were z-transformed. Variance (σ^2) of the random effect, bird individual ($n = 64$), was 0.363 and 0.275 for intertidal and total foraging time model, respectively

		Intertidal foraging time					Total foraging time				
		Estimate	SE	df	χ^2	sig.	Estimate	SE	df	χ^2	Sig.
Environment	Intercept	3.26	0.21				4.89	0.21			
	Winter	—					—				
	Seasonday	-0.14	0.03	1	25.6	***	-0.18	0.03	1	37.8	***
	Seasonday ²	-0.19	0.02	1	64.5	***	-0.22	0.02	1	87.6	***
	Temperature	-0.18	0.04				-0.16	0.04			
	Temperature ²	—					—				
	Windspeed	0.13	0.02	1	60.4	***	0.17	0.02	1	109.7	***
	Windspeed ²	—					0.05	0.01	1	16.2	***
	Exposure	7.47	0.35				6.36	0.35			
	Exposure ²	-5.17	0.41				-5.28	0.40			
	10-tide exposure	6.68	0.78				3.31	0.78			
	10-tide exposure ²	-8.39	0.84				-6.00	0.84			
	Night	-3.45	0.08				-3.52	0.08			
	Moon	-0.72	0.08				-0.98	0.08			
Feeding specialization	Night × moon	1.14	0.12				2.16	0.12			
	Bth	0.89	0.19				-0.90	0.19			
	Bth × night	0.60	0.08				0.64	0.08			
	Bth × moon	-0.07	0.08				0.16	0.08			
	Bth × night × moon	0.24	0.12	1	4.3	*	-0.71	0.12	1	37.8	***
	Bth × temperature	-0.07	0.02	1	8.9	**	-0.08	0.02	1	11.9	***
	Bth × exposure	-0.77	0.26	1	9.0	**	1.33	0.26	1	27.1	***
	Bth × exposure ²	0.19	0.29	1	0.4	ns	-0.71	0.72	1	10.4	**
	Bth × 10-tide exposure	-4.23	0.72	1	34.4	***	1.46	0.77	1	1.0	ns
	Bth × 10-tide exposure ²	4.05	0.77	1	27.7	***	-0.92	0.29	1	3.6	ns

Sig. = *0.05 > P > 0.01, **0.01 > P > 0.001, *** P < 0.001.

Differential susceptibility to environmental change

Worm specialists complement intertidal foraging with grassland foraging when intertidal flat exposure is limited by high water levels (driven by the lunar cycle or wind). By utilizing grasslands, worm specialists maintain their foraging time under conditions of high water levels. Shellfish specialists may lack the specialization to feed on grasslands and, hence, do not feed on grasslands when intertidal flat exposure is limited. To compensate for extended high water levels and to avoid a deteriorating body condition, they would need to either prolong intertidal foraging time in subsequent tides or increase foraging efficiency. We found no distinct relation between shellfish specialist foraging time and average intertidal exposure time over the last ten tides, suggesting that shellfish specialists do not compensate in subsequent tides. This would indicate that shellfish specialists may increase their foraging efficiency, for example, by capturing larger prey and by handling prey faster, but this comes at the cost of an increased risk of bill damage (Swennen et al. 1989; Rutten et al. 2006). This can explain why foraging efficiency on intertidal flats does not per se differ among feeding specializations under normal environmental conditions (Boates and Goss-Custard 1989; Boates and Goss-Custard 1992; Durell et al. 1997). It is likely that worm specialists maximize intertidal foraging efficiency under all conditions as capturing and eating soft-bodied prey does not require specific techniques nor involves risk of bill damage (Durell et al. 1997). Shellfish specialists, however, may only increase their foraging efficiency when intertidal flat accessibility is limited. Winters with prolonged periods of high water levels may, therefore, increase mortality risk of shellfish

specialists. These winters are more likely to occur in the future with ongoing sea-level rise due to climate change (Rahmstorf 2007) and more localized effects, such as soil subsidence, due to mining.

Worm specialists and shellfish specialists showed contrasting trends in foraging time with temperature. As expected, shellfish specialists increased their foraging time when it was colder, probably, to achieve a higher food intake that is needed due to higher metabolic rates and energy expenditure associated with colder temperatures (Kersten and Piersma 1987). An important constraint for worm specialists in cold weather conditions is that they cannot utilize grasslands as worms and invertebrates are unreachable in frozen soil. Also, soft-bodied preys on intertidal flats are more difficult to obtain under colder conditions than shellfish as they become less active (Esselink and Zwartz 1989; Zwartz and Wanink 1993). Consequently, worm specialists may be more vulnerable to cold spells as their food intake depends on intertidal foraging only. Indeed, contrary to what would be expected given the increased energy expenditure required in cold temperatures, worm specialists shorten instead of lengthen foraging time when it is cold. This suggests that foraging time is limited for worm specialists in cold weather. Our results are consistent with previous studies, which found that worm specialists suffer from a higher mortality than shellfish specialists during winters with cold spells (Durell et al. 1993, 2001; Durell 2007), but it does not help us understand why a later study found the opposite trend (van de Pol et al. 2010). These results are relevant in the context of climate change as specializing in worms is expected to become a less risky strategy in the future as cold winters may be less likely to occur.

Allocation of foraging time over daytime and nighttime tides differed among feeding specializations and also depended on the moon stage. During full moon, more foraging time was allocated to nocturnal tides (see also [Heppleston 1971](#)), probably, because foraging efficiency increases in illuminated conditions compared to dark conditions ([Santos et al. 2010](#)). Interestingly, grassland foraging by worm specialists occurred mainly during nighttime full moon tides, suggesting that under these conditions worm specialists can realize an intake on inland grasslands, which is similar or higher than intake on intertidal flats. Many insects and worms adjust their life cycle to the moon phase ([Raible et al. 2017](#)), which has been proposed as a potential explanation for full moon grassland foraging by birds ([Ydenberg et al. 1984](#)). All birds allocated more foraging time to daytime tides, probably, because waders achieve a higher foraging efficiency in daylight ([Zwarts and Drent 1981](#); [Lourengo et al. 2008](#)). We found that worm specialists allocated approximately 1–1.4 h (depending on moon stage) more foraging time to daytime tides compared with shellfish specialists. This suggests that the difference in foraging efficiency between day and night is more pronounced in worm specialists, probably, because they locate prey mostly visually ([Ens et al. 1996](#)). The increased reliance on daytime foraging may increase the vulnerability of worm specialists to daytime disturbance in foraging areas compared to shellfish specialists. Upon disturbance, available foraging time is shortened and energetic costs increase, which is especially costly if individuals have to compensate by foraging at times when foraging efficiency is low as might be the case for worm specialists during the night. Consequently, we hypothesize that critical disturbance thresholds, the frequency of disturbance at which animals lose condition, are not only dependent on environmental conditions ([Goss-Custard et al. 2006](#)) but also differ between individuals with different feeding strategies.

Phenotypic plasticity in feeding specialization is a potential mechanism via which oystercatchers anticipate changes in food availability and environmental conditions. However, individuals often consistently consume the same diet over multiple winter seasons, suggesting that their flexibility is limited (Goss-Custard and Durell 1983). The ability to switch between different feeding specializations is likely constrained by bill morphology. For example, the longer and slender bill of females can reach more deeply buried prey, such as worms and thin-shelled shellfish, but is less suitable to break into the armor of thick-shelled shellfish in comparison with the shorter and higher bill of males (Hulscher and Ens 1992). Although feeding specializations often remain consistent over winters, individuals can change their diet and bill tip shape between seasons. For example, diet switches from soft-bodied prey in summer to shellfish in early winter are documented (Boates and Goss-Custard 1989). Moreover, shellfish feeders that breed inland switch to a worm diet upon arrival in the breeding area (inducing a pointy bill shape change) and back to a shellfish diet upon arrival in the wintering area in autumn (Hulscher 1984). Although bill shape changes can take place within several weeks (Hulscher 1984), diet switches from soft-bodied prey to shellfish are likely costly because they involve a high risk of bill damage (Rutten et al. 2006). Tracking individuals over multiple years can reveal the extent to which feeding specialization and associated foraging time allocation strategies develop or remain similar over time.

Broader implications

By quantifying spatiotemporal foraging time using combined GPS and accelerometer measurements, we provide important insights

in the behavioral mechanisms of individuals with different feeding specializations in response to environmental conditions. In general, foraging time estimates, as quantified in this study, can be used to validate simulated foraging times in population models to determine whether virtual individuals behave similar to real individuals. Although those models ultimately focus on how body condition, survival, or population numbers are affected, foraging time can also be derived for validation in individual-based models where individuals make decisions to forage during every time step (e.g., [Stillman and Goss-Custard 2010](#)). Moreover, based on behavioral adaptations, we can make predictions on how vulnerable individuals are to specific environmental conditions. Being able to measure environmental susceptibility in relatively short-term studies is important as measuring actual fitness consequences often require long-term studies. For example, extremely cold winters rarely occur in Western Europe and, it has been shown that, even in studies lasting multiple decades, it is hard to reliably estimate how feeding specialization in oystercatchers moderates susceptibility of adult survival to cold weather ([van de Pol et al. 2010](#); [Bailey and van de Pol 2016](#)). Implementing behavioral responses to environmental conditions in population models may be used as a shortcut to predict responses of different feeding specializations to environmental change.

SUPPLEMENTARY MATERIAL

Supplementary data are available at *Behavioral Ecology* online.

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Data accessibility: Analyses reported in this article can be reproduced using the data provided by [van der Kolk et al. \(2019\)](#).

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